

Lecture 6: Software & Operating Systems

System Software · Operating Systems · Application Software · Windows & Linux Overview

Learning Outcomes

- Distinguish between hardware and software, and explain why a computer needs both to function.
- Define system software and application software, and differentiate between them with examples.
- Define an operating system and explain, in detail, each of its major functions.
- Describe the main types of operating systems, with examples of where each is used.
- Differentiate between general-purpose and specific-purpose application software.
- Describe the key features of Windows and of Linux, and explain the meaning of “open source”.
- Compare Windows and Linux across multiple parameters, and recommend the right choice for a given type of user.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations, diagrams, and examples.

1 Recap — From Hardware to Software

Lectures 3 to 5 studied a computer's physical parts in detail — the block diagram, memory and storage, and input/output devices. All of that hardware, on its own, cannot do anything useful by itself. It needs software: organised sets of instructions that tell the hardware exactly what to do. This lecture studies software itself, starting with the most important piece of software on any computer — the operating system — and finishes by comparing the two operating system families you are most likely to encounter: Windows and Linux.

Hardware vs Software

Hardware is the physical, touchable parts of a computer — the keyboard, monitor, CPU, RAM. Software is an intangible set of instructions (a program) that tells that hardware what to do. Neither is useful without the other: hardware provides the raw processing power, and software directs exactly how that power is used.

2 Types of Software — Overview

Software is broadly divided into two categories, based on who it primarily serves: the computer itself, or the user.

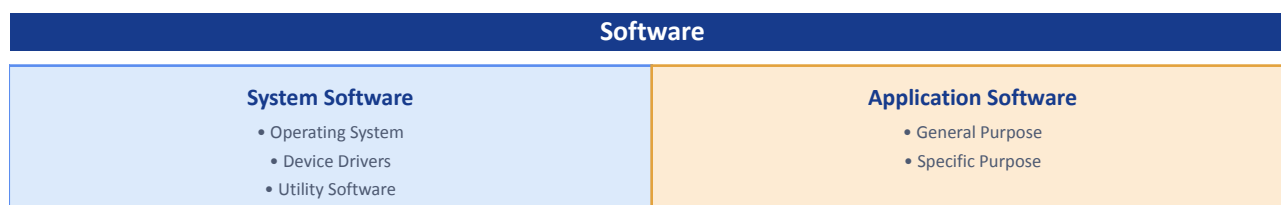


Fig 6.1 — The two broad categories of software.

Software is a collection of **programs, instructions, and data** that tells a computer what to do and how to perform a particular task. Unlike hardware, software **cannot be physically touched**.

For example, **Windows, MS Word, Google Chrome, Paint, and VLC Media Player** are software.

Software is mainly divided into **two types**:

1. System Software

System software is a type of software that controls and manages the **hardware and basic operations of a computer**. It also provides a platform for running application software.

Examples: Windows, Linux, macOS, device drivers, etc.

Main types of System Software:

- **Operating System:** Manages computer hardware and software resources.
Examples: Windows, Linux, macOS.
- **Device Drivers:** Help the operating system communicate with hardware devices.
Examples: Printer driver, graphics driver.
- **Utility Software:** Helps maintain, protect, and improve the performance of a computer.
Examples: Antivirus, backup software, disk cleanup tools.
- **Language Translators:** Convert programming languages into a form that the computer can understand.
Examples: Compiler, Interpreter, Assembler.

2. Application Software

Application software is designed to help users perform **specific tasks**.

Examples:

- **MS Word** – Creating documents
- **MS Excel** – Calculations and spreadsheets
- **PowerPoint** – Creating presentations
- **Google Chrome** – Browsing the Internet
- **VLC Media Player** – Playing audio and video
- **Photoshop** – Editing images

Application software can be further classified as:

- **General-Purpose Software:** Used for common tasks by many users, such as Word, Excel and web browsers.
- **Special/Customized Software:** Developed for a particular organization or specific task, such as **payroll, billing, hospital or college management software**.

Difference Between System Software and Application Software

System Software	Application Software
Manages and controls the computer	Performs specific user tasks
Works mainly in the background	Directly used by the user
Essential for basic computer operation	Installed according to user requirements
Example: Windows, Linux	Example: MS Word, Excel

3 System Software

System software is a category of software that manages and controls a computer's hardware, and provides a platform on which application software can run. The user rarely interacts with system software directly — it works quietly in the background, making the rest of the computer usable.

4 Operating Systems

An operating system (OS) is system software that acts as an intermediary between the user, the application software they run, and the computer's hardware. It manages every hardware resource — the CPU, memory, storage, and input/output devices — and provides application software with a common, simplified way to use them.

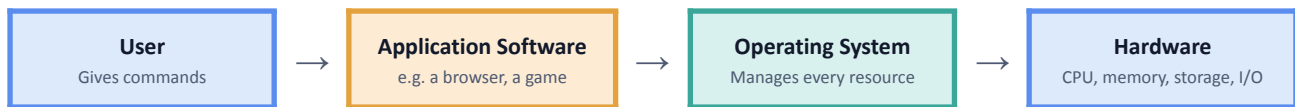


Fig 6.2 — The operating system sits between the user/application software and the hardware.

Important Point for Exams

No application software can talk to hardware directly — every request, from saving a file to printing a page, is routed through the operating system. This is why the OS is often called the most important piece of software on any computer.

Functions of an Operating System

Operating System				
Process Management Schedules CPU time between programs	Memory Management Allocates RAM to running programs	File Management Organises files & folders on storage	Device Management Coordinates input/output devices	Security & Interface Protects data; provides CLI/GUI

Fig 6.3 — The five major functions of an operating system.

Function	Explanation
Process Management	Creates, schedules, and terminates processes (running programs), allocating CPU time between them so multiple programs appear to run at once.
Memory Management	Allocates and frees memory space for each running program, and keeps track of which parts of RAM are currently in use.
File Management	Organises, names, stores, retrieves, and protects files and folders on secondary storage devices.
Device Management	Manages input and output devices through device drivers, coordinating requests from programs to use hardware such as a printer, keyboard, or disk.
Security & Access Control	Protects data and resources from unauthorised access, managing user accounts, passwords, and permissions.
User Interface	Provides a way for the user to interact with the computer — either a Command Line Interface (CLI), where the user types text commands, or a Graphical User Interface (GUI), which uses windows, icons, and menus.

Types of Operating Systems

Type	Description
Batch OS	Processes a queue of jobs one after another with no user interaction during execution; used on early mainframes.
Time-Sharing OS	Allows multiple users to use the same computer at once, by rapidly switching CPU time between them so each feels they have full attention.
Real-Time OS (RTOS)	Guarantees that a task is completed within a strict, predictable time limit; used in embedded systems, medical devices, and industrial control.
Multi-user / Multi-tasking OS	Supports multiple users and/or multiple programs running seemingly simultaneously; used by most modern desktop and server operating systems.
Distributed OS	Manages a group of independent, networked computers so that they appear to the user as a single system.

5 Application Software

Application software is software designed to help the user perform a specific task, rather than manage the computer itself. Unlike system software, the user chooses when to open and close it.

Type	Description	Examples
General Purpose	Usable for a wide variety of common tasks by almost any user.	MS Word, MS Excel, a web browser, a media player
Specific Purpose (Customised)	Built for one particular task or organisation, and not generally useful outside it.	Payroll software, a hospital management system, railway reservation software
System Software vs Application Software — A Frequently Asked Differentiate Question Refer back to the comparison table in Section 2 (Fig 6.1) for the full system-vs-application comparison — examiners very frequently ask this exact question, so make sure you can reproduce that table confidently.		

6 Overview of Windows

Windows is a **Graphical User Interface (GUI) based operating system** developed by **Microsoft**. An operating system is system software that acts as an **interface between the user and computer hardware** and manages the resources of the computer.

Windows allows users to interact with the computer using **windows, icons, menus, buttons, and a mouse pointer**, making it easy to use.

Main Features of Windows Operating System

1. **Graphical User Interface (GUI):**
Windows provides an easy-to-use graphical interface with icons, menus, windows, and buttons.
2. **Desktop:**
The **Desktop** is the main screen that appears after Windows starts. It contains icons, shortcuts, the taskbar, and other items.
3. **Start Menu:**
The Start Menu provides access to **applications, settings, files, search, and power options**.

4. Taskbar:

The Taskbar is generally located at the bottom of the screen. It displays the **Start button, running applications, pinned programs, and notification area.**

5. File Explorer:

File Explorer is used to **view, organize, copy, move, rename, and delete files and folders.**

6. Multitasking:

Windows allows users to run **multiple applications at the same time.** For example, a user can work in MS Word while listening to music.

7. Hardware Support:

Windows supports various hardware devices such as **printers, keyboards, mice, scanners, cameras, and external storage devices.**

8. User Accounts and Security:

Windows provides features such as **user accounts, passwords, Windows Security, firewall, and system updates** to protect the computer.

Common Versions of Windows

Some popular versions are **Windows XP, Windows 7, Windows 8, Windows 10, and Windows 11.**

Basic Elements of Windows

The main elements of the Windows screen include:

- Desktop
- Icons
- Start Menu
- Taskbar
- File Explorer
- Recycle Bin
- Notification Area

Feature	Details
Source code	Proprietary (closed source) — owned by Microsoft and not publicly available.
Cost	Requires a paid licence to use legally.
Interface	Graphical User Interface (GUI) by default, with a desktop, taskbar, and File Explorer.
Compatibility	Extremely wide range of compatible hardware and commercial software, including most popular games.
File system	Primarily NTFS (New Technology File System).
Popular versions	Windows XP, Windows 7, Windows 10, Windows 11.

7 Overview of Linux

Linux is a **free and open-source operating system** based on Unix. It manages the hardware and software resources of a computer and provides an interface between the **user and computer hardware.**

Linux was originally created by **Linus Torvalds** in **1991**. Today, it is widely used in **personal computers, servers, supercomputers, embedded systems, and cloud computing**.

Main Features of Linux Operating System

1. **Free and Open Source:**
Linux is freely available, and its source code can be viewed and modified according to the applicable software licence.
2. **Multiuser:**
Linux allows **multiple users** to use the same computer system while maintaining separate accounts and permissions.
3. **Multitasking:**
It can run **multiple programs at the same time**.
4. **Security:**
Linux provides security features such as **user accounts, passwords, file permissions, and access control**.
5. **Stable and Reliable:**
Linux is known for its stability and is widely used on **servers** that need to operate continuously.
6. **Command-Line Interface:**
Linux provides a powerful **terminal or command-line interface (CLI)** through which users can perform various operations using commands.
7. **Graphical User Interface:**
Linux can also provide a graphical desktop environment with **windows, icons, menus, and a mouse pointer**.
8. **Portable:**
Linux can run on many different types of hardware, from small embedded devices to powerful servers and supercomputers.

Popular Linux Distributions

Different versions or packages of Linux are called **Linux distributions (distros)**. Some popular distributions are:

- Ubuntu
- Debian
- Fedora
- Linux Mint
- Red Hat Enterprise Linux

Uses of Linux

Linux is commonly used in **web servers, cloud systems, supercomputers, software development, cybersecurity, networking, and embedded devices**.

Feature	Details
Source code	Open source — freely available for anyone to view, modify, and redistribute.
Cost	Most distributions are free to download and use.
Interface	Can be operated through a GUI or a Command Line Interface (CLI); many servers run with no GUI at all.
Customisation	Highly customisable — almost every part of the system can be changed or replaced.

Feature	Details
File system	Commonly ext4.
Popular distributions	Ubuntu, Fedora, Debian, Red Hat, Linux Mint.

What is a “distribution”? Since Linux is open source, different organisations package the Linux kernel together with their own choice of software and tools to create a complete, ready-to-install operating system. Each such package is called a distribution, or “distro” for short — Ubuntu and Fedora are two of the most popular examples.

Real-Life Example

You are almost certainly using a Linux-based device right now without realising it: Android, the operating system on most smartphones, is built on a modified Linux kernel. Linux also powers the vast majority of the world's web servers and supercomputers.

8 Windows vs Linux — Full Comparison

Point	Windows	Linux
Source code	Closed (proprietary)	Open source
Cost	Paid licence required	Mostly free
Customisation	Limited	Highly customisable
Security	A common target, due to its popularity	Generally considered more secure, less commonly targeted
Ease of use for beginners	Very user-friendly, minimal setup	Can have a steeper learning curve, though modern distros are much friendlier
Software availability	Very wide, especially commercial software and games	Wide for servers and development tools; fewer mainstream commercial desktop apps and games
Typical users	Home users, offices, gamers	Developers, server administrators, researchers
File system	NTFS	ext4
Example	Windows 10, Windows 11	Ubuntu, Fedora, Debian

Important Point for Exams

Neither Windows nor Linux is universally “better” — the right choice depends entirely on the user's needs. Examiners often ask you to recommend an OS for a specific scenario, so always justify your answer with the relevant feature from this table, rather than just naming an OS.

9 Putting It All Together — A Worked Example

Consider what happens, in terms of software, from the moment a computer is switched on to a user saving a document:

Step	Software Involved	What Happens
1	ROM (BIOS/UEFI firmware)	The computer's start-up firmware, stored in ROM, runs first and locates the operating system on the storage drive.
2	Operating System	The OS (Windows or Linux) loads into RAM, initialises the hardware, and presents the user with a desktop or command line.
3	Application Software	The user opens a word processor — application software running on top of the OS, using its services to access memory, storage, and the display.
4	OS — Process & Memory Management	As the user also opens a web browser, the OS allocates CPU time and RAM to both programs so they appear to run at the same time.
5	OS — File Management	When the user presses Save, the operating system's file management function writes the document to the storage drive.

This everyday sequence uses almost every idea in this lecture: system software (the OS and its functions) working invisibly in the background, and application software directly serving the user — all built on the ROM, memory, and storage studied in earlier lectures.

10 Fill in the Blanks

- 1. Software that manages computer hardware and provides a platform for other software to run is called _____ software.
- 2. Software designed to help the user perform a specific task, such as word processing, is called _____ software.
- 3. The operating system function that allocates CPU time among multiple running programs is called _____ management.
- 4. The operating system function that organises and stores files on a disk is called _____ management.
- 5. A(n) _____ interface allows the user to interact by typing text commands, while a(n) _____ interface uses windows, icons, and menus.
- 6. Windows was developed by the company _____.
- 7. Linux is based on the design principles of an earlier operating system called _____.
- 8. Unlike Windows, Linux's source code is _____, meaning it can be freely viewed, modified, and redistributed.
- 9. A packaged, ready-to-install version of Linux, such as Ubuntu or Fedora, is called a _____.
- 10. The Windows file system is called _____, while a common Linux file system is called _____.

Answers

1. System | 2. Application | 3. Process | 4. File | 5. Command Line (CLI), Graphical User (GUI) | 6. Microsoft | 7. UNIX | 8. Open source | 9. Distribution (distro) | 10. NTFS, ext4

11 True or False

- 1. System software is designed to help users perform specific tasks like writing a document.
- 2. The operating system manages hardware resources such as memory and storage devices.
- 3. A word processor is an example of system software.
- 4. Linux is open source, meaning its source code is freely available to view and modify.
- 5. Windows is generally more customisable than Linux.
- 6. Android smartphones run on a modified version of the Linux kernel.
- 7. A Real-Time Operating System guarantees that tasks are completed within a strict time limit.
- 8. Application software can run without any operating system installed.

Answers with Explanation

- 1. False.** It is the reverse: application software helps users perform specific tasks; system software manages the computer itself.
- 2. True.** Managing hardware resources such as memory, storage, and devices is a core function of the operating system.
- 3. False.** A word processor is application software, not system software.
- 4. True.** Open source is precisely Linux's defining characteristic — anyone can view, modify, and redistribute its code.
- 5. False.** It is the reverse: Linux is generally far more customisable than Windows.
- 6. True.** Android is built on a modified Linux kernel.
- 7. True.** That is the defining feature of a Real-Time Operating System.
- 8. False.** Application software depends on the operating system to access hardware and run; it cannot function without one.

12 Short Answer Questions (2–3 Marks)

- Differentiate between system software and application software. (3 marks)
- What is an operating system? State two of its functions. (3 marks)
- Name any four functions of an operating system. (2 marks)
- Differentiate between a CLI and a GUI. (2 marks)
- What is a Real-Time Operating System? Give one example use. (2 marks)
- Give two examples each of general-purpose and specific-purpose application software. (2 marks)
- What does it mean to say that Linux is “open source”? (2 marks)
- Name any two popular Linux distributions. (2 marks)
- State two differences between Windows and Linux. (3 marks)

13 Long Answer Model Answers (5, 10 & 15 Marks)

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

Q1 (5 Marks): Differentiate between system software and application software, with examples.

System software is software that manages and controls a computer's hardware, and provides a platform on which other software can run. It runs automatically in the background from the moment the computer starts, and the user rarely interacts with it directly. The operating system, device drivers, and utility software are all examples of system software.

Application software, in contrast, is software designed to help the user perform a specific task, such as writing a document, browsing the web, or playing a game. Unlike system software, it only runs when the user chooses to open it, and it depends entirely on system software — particularly the operating system — to access the computer's hardware. MS Word, a web browser, and a media player are all examples of application software. In short, system software serves the computer itself, while application software serves the user directly.

Q2 (10 Marks): Explain the functions of an operating system in detail.

An operating system is system software that acts as an intermediary between the user, application software, and the computer's hardware, managing every hardware resource on the user's behalf. It performs several distinct functions.

Function	Explanation
Process Management	Creates, schedules, and terminates processes, allocating CPU time between them so multiple programs appear to run at once.
Memory Management	Allocates and frees memory space for each running program, tracking which parts of RAM are currently in use.
File Management	Organises, names, stores, retrieves, and protects files and folders on secondary storage.
Device Management	Manages input and output devices through device drivers, coordinating requests to use hardware such as a printer or disk.
Security & Access Control	Protects data and resources from unauthorised access, managing user accounts and permissions.
User Interface	Provides a Command Line Interface (CLI) or Graphical User Interface (GUI) for the user to interact with the computer.

In conclusion, without these functions working together continuously, no application software could safely or efficiently use the computer's hardware — which is exactly why the operating system is considered the single most important piece of software on any computer.

Q3 (10 Marks): Give an overview of Windows and Linux operating systems, highlighting their key features.

Windows is an operating system developed by Microsoft, first released in 1985, and remains one of the most widely used operating systems for personal computers today. Its source code is proprietary (closed), meaning it is owned by Microsoft and not publicly available, and it requires a paid licence to use legally. Windows is known for its user-friendly Graphical User Interface, extremely wide compatibility with commercial software and hardware (including most games), and its primary file system, NTFS.

Linux is an operating system originally created by Linus Torvalds in 1991, based on design principles borrowed from the older UNIX operating system. Unlike Windows, Linux is open source: its code is freely available for anyone to view, modify, and redistribute, and most versions are free to use. Because it is open source, different organisations package the Linux kernel with their own choice of software to create complete, installable operating systems called distributions, such as Ubuntu, Fedora, and Debian. Linux can be operated through a GUI or a Command Line Interface, is highly customisable, and commonly uses the ext4 file system. It is especially popular for servers, software

development, and embedded devices — Android, the operating system used on most smartphones, is itself built on a modified Linux kernel.

In conclusion, Windows and Linux represent two very different philosophies: Windows prioritises ease of use and broad commercial compatibility behind closed, paid software, while Linux prioritises openness, flexibility, and free access to its underlying code.

Q4 (15 Marks): Compare Windows and Linux in detail across multiple parameters, and discuss which one would be more suitable for different types of users, with examples.

Windows and Linux differ across almost every parameter that matters to a user choosing an operating system.

Point	Windows	Linux
Source code	Closed (proprietary)	Open source
Cost	Paid licence required	Mostly free
Customisation	Limited	Highly customisable
Security	A common target, due to popularity	Generally more secure, less targeted
Software availability	Very wide, especially commercial software and games	Wide for servers and development tools; fewer commercial desktop apps and games
Typical users	Home users, offices, gamers	Developers, server administrators, researchers

A home user or gamer would generally be better served by Windows, since it offers the widest compatibility with commercial games and everyday software, along with a simple, familiar graphical interface that requires little technical knowledge to use.

A software developer often prefers Linux, since its open-source nature, powerful command-line tools, and close relationship with the software running on most servers make it a natural environment for writing and testing code intended to run on those same servers.

A server administrator managing a company's web servers would very likely choose Linux, since it is free to deploy at scale, highly customisable to the exact needs of the server, and widely regarded as stable and secure for long-running, unattended systems — which is precisely why the majority of the world's web servers run on Linux.

In conclusion, there is no single “better” operating system in absolute terms: Windows suits users who prioritise ease of use and broad commercial software compatibility, while Linux suits users who prioritise cost, customisation, and control — and the right choice always depends on matching these features to the specific needs of the user.

14 Exam Tips

- Memory trick for OS functions: “PM-FDS” — Process, Memory, File, Device management, plus Security and User interface.
- Memory trick for the OS's position: “User → Application → OS → Hardware” — the OS always sits between application software and hardware, never the other way round.
- For “differentiate” questions (system vs application software, Windows vs Linux, CLI vs GUI), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- Always name at least one real example when discussing Windows or Linux (Windows 11; Ubuntu, Fedora) — concrete examples earn easy marks.
- Common Mistake: Do not say application software can run without an operating system — it always depends on the OS to access hardware.

- Common Mistake: Do not say Linux is only used by programmers — mention Android and web servers as major everyday uses too.
- For 10 and 15 mark questions, structure your answer with a clear paragraph or table per component, exactly like the model answers in Section 13 — examiners reward clear structure over long, unstructured paragraphs.

Lecture 6 — Software & Operating Systems — bringing the hardware from Lectures 3–5 to life.